

STAG: A Tool for realtime Replay and Analysis of Spatial Trajectory and Gaze Information captured in Immersive Environments

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ABSTRACT

This article describes a toolkit called S.T.A.G for real-time capture, playback, and further analysis of user spatial trajectory and gaze information. We developed S.T.A.G as part of our ongoing investigation of spatial trajectory analysis by capturing real-time user behavior inside a maze-like immersive environment. S.T.A.G can be generally applied to any immersive environment setup and is meant to serve as a toolkit providing deeper insights into user spatial trajectory analyses. The captured user activity log consists of its entire spatial trajectory set and its gaze information captured at an interval of 10 milliseconds per frame. We implemented S.T.A.G using C# and deployed it using the Unity™ real-time game engine.

Index Terms: Virtual Reality—Analysis—Spatial Trajectory—Gaze Information

1 INTRODUCTION

Navigating spaces is an embodied experience. Most human activities involve a fair bit of navigating physical and, more recently, virtual spaces. However, it is not unsafe to assume that in the case of modern mobile, untethered virtual reality setups, the cognitive functions of the user would dominate the motor functions. This assumption is reasonably valid, primarily because a majority of mobile virtual reality experiences rely on controller-based input from the user to navigate the immersive content. With the on-going advances in the computing fields, such as visualization and augmented/mixed/virtual reality (A/M/VR), there is a considerable uptick in end-user participation specifically in the entertainment and video-gaming sector. This upward trend of VR usability naturally begs the question of standardizing the usability data collection and analyses apparatus. The premise of our work started with questioning the above assumptions and as a side effect S.T.A.G was realized following the footsteps of another similar toolkit developed earlier by Steed et al. [1].

2 BACKGROUND APPLICATION FRAMEWORK

Our maze application has been a testbed for a series of user analytics experimentation over several years. The first iteration drew upon the works of Basu et al. design [2], which showed us how to layer an immersive experience with the backdrop of a palatable story. In this article, we will be presenting two variations of our maze testbed application.

2.1 Environmental factors maze application

In this iteration, we investigate immersive spatial navigational performance to study the correlation between the physical and the virtual environment (indoors versus outdoors). This variation of the maze application required the user to traverse two sets of virtual environments resembling simulated indoor and outdoor setting while shooting virtual bull-eye targets. We suspect that the environment,

both physical and simulated, affects the overall spatial navigation performance involved in an immersive virtual experience.

2.2 Technical factors maze application

Our other maze application investigates immersive spatial navigational behaviorism based on the user's exposure to video-gaming principles and mechanics. This iteration requires the user to navigate a maze and find multiple target sites. For consistency, every player had a fixed starting point and fixed targets to reach in the maze. The user repeats two back to back sessions of the same maze but under two different 3D user interface (3DUI) schematics, namely immersive and non-immersive setting. The task model is loosely representative of an empathic setting, giving the user an impression of a rescuer, and the minimap is a reference to a minimalist context building.

3 SYSTEM HARDWARE AND SOFTWARE

Both the virtual reality maze experiences were developed using the Unity game engine and deployed on a smartphone enabled HMD device as mobile apps (both iOS as well as Android). While each participant was busy solving the maze, the system software kept an active log of their in-session trajectory data along with their head orientation data every 10 milliseconds. Furthermore, the system software also logged relevant event data such as participants quitting the maze experience or participants finding the targets successfully. Visual examples of participant tracking algorithm is illustrated in Figure 2 and Figure 3.

Once the user data was collected, scripts developed in C# programming language were used to analyze the users' trajectory data in Unity game engine. These scripts parse through user transformation (x,y,z) in real-time and visualize their search patterns in the maze. This, in turn, allows to confirm visually emerging differences in user performance in immersive virtual reality. This sort of visual analysis can be generalized to apply to other studies concerning spatial navigation in immersive virtual reality.

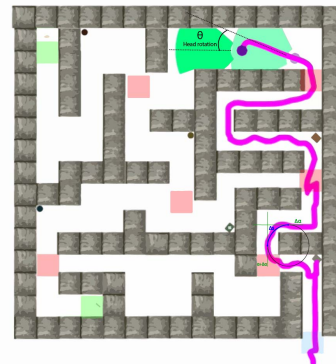


Figure 1: Technical factors study - Visualization of two trajectory features: positional curvature and head rotation amount.

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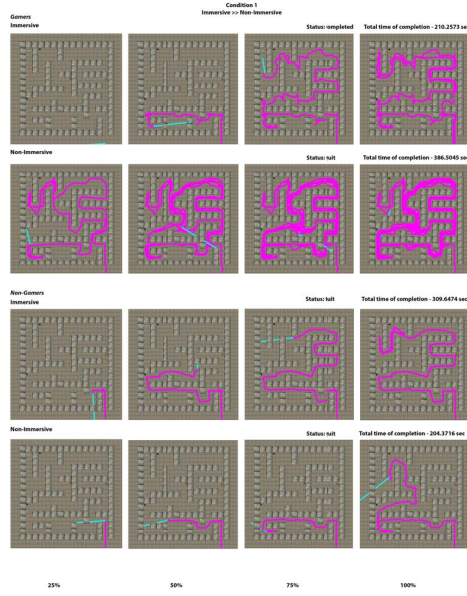


Figure 2: Technical factors study - Visualization of a randomly selected gamer and a non-gamer's spatial trajectory data for both sessions under Condition 1 at 25% increments of individual time of completion.

Features	Definition
Distance traveled	It is the total length of the path traveled between two positions.
Coverage	Total number of unit cubes covered (area).
Number of decision points reached	Total number of decision points covered in the maze.
Positional Curvature	The signed angle of curvature between two consecutive position vectors.
Head rotation amount	The unsigned head rotation angle between two consecutive rotation transforms.

Table 1: Technical factors study - Features extracted from users' trajectory data for deep exploration and their corresponding definitions.

4 MEASURES

Data automatically recorded during the course of the maze navigation was used to measure the performance of each participant. For each session, we recorded, every 10 millisecond, the participant's spatial trajectory inside the maze along with orientation of gaze information (see Figure 2 and Figure 3). This allowed us to effectively playback and calculate each participant's spatio-temporal activity including, but not limited to, time spent at decision nodes in the

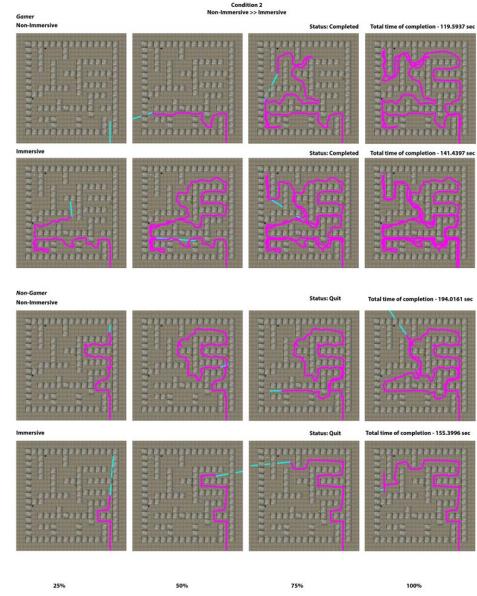


Figure 3: Technical factors study - Visualization of a randomly selected gamer and a non-gamer's spatial trajectory data for both sessions under Condition 2 at 25% increments of individual time of completion.

maze and cumulative gaze rotation at decision nodes in the maze. Additionally, for deep exploration of the trajectory data, we defined a set of mathematically derived features generalized for each trajectory such as distance traveled, decision points (nodes) reached inside the maze, positional curvature, head rotation amount, and coverage of the maze. Some of these trajectory features are highlighted in Table 1 as well as Figure 1. Positional curvature feature refers to the curvature of the trajectory calculated per frame between successive position vectors. Rotation amount feature refers to the unsigned angle calculated per frame between successive head rotation transforms stored as Quaternions. Coverage feature is simply the number of unit cubes covered (area) by the user, calculated per frame.

5 FUTURE WORK

We intend to expand on the set of trajectory feature definitions in our toolkit's future iteration. On that note, we are currently working on integrating Long-Short term Neural Network (LSTM) architecture with S.T.A.G to help expand hidden feature extraction and discovery in user trajectory datasets. Finally, we want to publish the S.T.A.G toolkit for the public/research/scholarly use through online repositories such as GitHub.

6 CONCLUSIONS

In essence, we have created a trajectory analysis toolkit for VR studies that lets us playback and analyze user trajectories for a deep insight into VR usability. We hope that the S.T.A.G toolkit will be helpful for a broad swath of VR experimental design that involves a lion-share of user trajectory analysis, specifically for people studying hippocampal activity mapping in rats (and/or perhaps humans) placed in mazes.

REFERENCES

- [1] A. Murgia, R. Wolff, W. Steptoe, P. Sharkey, D. Roberts, E. Guimaraes, A. Steed, and J. Rae, "A tool for replay and analysis of gaze-enhanced multiparty sessions captured in immersive collaborative environments," in *2008 12th IEEE/ACM International Symposium on Distributed Simulation and Real-Time Applications*, pp. 252–258, IEEE, 2008.
- [2] A. Basu, K. Johnsen, and K. Bogert, "Poster: Evolution and usability of ubiquitous immersive 3d interfaces," in *2013 IEEE Symposium on 3D User Interfaces (3DUI)*, pp. 131–132, March 2013.